

Original heat control and complete source endpoint

Split-Zero research programme

19 September 2026

This edition receives the latest supplied research continuation in full mathematical form. The first part proves subexponential growth of the original heat-comparison constant, evaluates the resulting response certification depth, and keeps all three polynomial current products and the source-valued arithmetic boundary. The second part evaluates the finite Beurling quotient and its exact map to the reconstructed source, then strengthens the endpoint estimates by keeping the complete finite regular correction and both cross terms. All primary multiplicities, original physical coordinates and minimizing maps are specified below.

The new refinements include a full quartet Schur error bound, every single-primary nilpotent metric scale, and the first critical determinant correction. They strengthen the existing native programme through its actual receiving maps; the remaining kernel allocation, proper-source target metric and individual projected signs are not assigned from these different quantities. Complete proof sources and human references are included in each part. The received response and cited programme sources accompany the edition.

Subexponential heat cost and the original arithmetic currents

We receive Sections 1–4 and 8 of the supplied continuation [1], prove their heat and current estimates in the original coordinates, and retain the distinction between the exact currents and the two specified approximations of them. The full root polynomial, physical measure, fixed period, original boundary vectors and both integration regions are retained. Equations HG1–20 also carry the new growth estimate into the preceding NP heat-depth and leakage receivers.

1 The original finite source and the exact growth constant

Use $D = -x\partial_x$, $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$ and the full divisor $h(w) = \prod_{\rho \in Z} (w - \rho)^{m_\rho}$ of $g(w) = 2\xi(w)$. Every m_ρ is the full multiplicity. Set $d = \deg h$, $F_h(y) = (g/h)(1/2 + iy)$ and $w_h(y) = |F_h(y)|^2/(2\pi)$. HM1–7 [2] constructs a_h with Mellin transform g/h and the exact divided-jet section $R_h u = r_h(u)(D)a_h$, where $j_h r_h(u) = j_h(h/g)u$. Thus

$$\mathcal{A}_N P = P(D)a_h, \quad \|\mathcal{A}_N P\|^2 = \int_{\mathbb{R}} |P(1/2 + iy)|^2 w_h(y) dy, \quad u = \mathsf{T}_h j_h P, \quad \mathsf{T}_h = j_h(g/h). \quad (\text{HG1})$$

The unit T_h includes every original divided derivative. For $N \geq d$, the full coefficient Gram and attained value form are

$$\mathbb{G}_N = \begin{pmatrix} A_0 & C_N \\ C_N^* & S_N \end{pmatrix}, \quad A_0 = R_h^* R_h, \quad H_N^{(1)} = A_0 - C_N S_N^{-1} C_N^*, \quad (\text{HG2})$$
$$c_N^2 = \lambda_{\max}((H_N^{(1)})^{-1/2} A_0 (H_N^{(1)})^{-1/2}).$$

The final identity follows directly from HM10: insert $A_0 = H_N^{(1)} + C_N S_N^{-1} C_N^*$ and take the largest eigenvalue. The attained form is the minimum of (HG1) at fixed u , over the complete relation ideal $h\mathcal{P}_{N-d}$. This is the quadratic minimum of [10, Appendix C.4.1], proved in HM by completing the square.

Theorem 1 (Subexponential growth with all density zeros retained). *On this fixed-divisor domain, $\log c_N = o_h(N)$ as $N \rightarrow \infty$. For each fixed admissible R , the explicit estimate (HG5) below holds.*

Proof. First consider the stipulated quartet. Its midpoint is not a zero of g : the convergent alternating series for $\eta(1/2)$ is a sum of positive consecutive pairs, hence positive, and $\zeta(1/2) = \eta(1/2)/(1 - \sqrt{2}) \neq 0$. Also $h(1/2) \neq 0$. Define $f_0 = |F_h(0)|$, $M_0 = \max_{|z|=1} |F_h(z)|$, $b_0 = \min(1/2, f_0/(4M_0))$ and $w_* = f_0^2/(8\pi)$. Cauchy's coefficient estimate gives $|F_h(y) - F_h(0)| \leq M_0 b_0/(1 - b_0) \leq f_0/2$ on $[-b_0, b_0]$; therefore $w_h \geq w_*$ there.

For Legendre polynomials the recurrence recorded in [8, Table 18.9.1] proves $|P_n(z)| \leq (1+2r)^n$ on $|z| \leq r$. Indeed the induction uses $2r/(1+2r) + 1/(1+2r)^2 \leq 1$, and the initial two polynomials obey the bound. Their squared norm on $[-1, 1]$ is $2/(2n+1)$. Expansion in the orthonormal basis on $[-b_0, b_0]$ and Cauchy-Schwarz consequently proves, with $R_h^{\text{root}} = \max_{h(\rho)=0} |(\rho - 1/2)/i|$ and $B_h = 1 + 2(R_h^{\text{root}} + 1)/b_0$,

$$\sup_{|z| \leq R_h^{\text{root}} + 1} |P(1/2 + iz)| \leq \frac{(N+1)B_h^N}{\sqrt{2b_0w_*}} \|\mathcal{A}_N P\|. \quad (\text{HG3})$$

Here $\sum_{n=0}^N (2n+1) = (N+1)^2$. Cauchy's formula on each unit circle about an original jet point bounds every divided derivative by the same supremum. The coordinate change $w = 1/2 + iz$ contributes only powers of i of modulus one, which are retained in j_h . Taking the minimum at fixed u yields

$$H_N^{(1)} \succeq \frac{2b_0w_*}{d\|\mathsf{T}_h\|^2(N+1)^2B_h^{2N}} I, \quad c_N \leq C_h(N+1)B_h^N, \quad C_h = \sqrt{\frac{d\|\mathsf{T}_h\|^2\|A_0\|}{2b_0w_*}}. \quad (\text{HG4})$$

For the sharper rate take $R > 0$ with neither endpoint a real zero of the entire function F_h . Its real zeros in this interval are finite. Let $p_R(y) = \prod_{F_h(t)=0, |t| \leq R} (y-t)^{n_t}$, with full analytic multiplicities, $d_R = \deg p_R$, and $\omega_R = \min_{[-R,R]} |F_h/p_R|^2/(2\pi) > 0$. Choose fixed small circles about the jet points on which F_h has no zero, possible because g/h is a unit at every full root of h . Let their radii be r_ρ , let R_0 bound their distance from zero in the y coordinate, and put $m_R = \min_{\text{circles}} |p_R| > 0$ and $D_h^2 = \sum_\rho \sum_{j < m_\rho} r_\rho^{-2j}$. The full density gives $\|p_R P(1/2 + i\cdot)\|_{L^2[-R,R]} \leq \omega_R^{-1/2} \|\mathcal{A}_N P\|$. Apply the preceding Legendre proof to this degree- $(N+d_R)$ polynomial on $[-R, R]$, and divide by p_R only on those circles. Cauchy's formula and the original T_h give

$$c_N \leq C_{h,R}(N+d_R+1)(1+2R_0/R)^{N+d_R}, \quad C_{h,R} = \frac{\sqrt{\|A_0\|} \|\mathsf{T}_h\| D_h}{m_R \sqrt{2R\omega_R}} < \infty. \quad (\text{HG5})$$

For each fixed R , divide its logarithm by N and let N tend to infinity. The upper limit is at most $\log(1+2R_0/R)$. There are arbitrarily large admissible R . Letting them tend to infinity proves the upper limit zero. Since $H_N^{(1)} \preceq A_0$, (HG2) gives $c_N \geq 1$ and the lower limit is at least zero. All constants depending on R were fixed before the N limit. $\square$

For any family of one-factor degrees at most $2q+2$ the same bound also proves $\log c_* = o_h(q)$, where $c_* \geq 1$ is their maximum. Indeed (HG5) bounds all of them by its value at $2q+2$ before taking the same successive limits. The preceding HM divergence $c_N \rightarrow \infty$ is compatible with this zero exponential rate.

2 Exact currents before approximating their metric

At the original cutoff retain the positive $G = G_N$, actual observation $\Lambda : E \rightarrow \text{im } \Lambda$, and

$$Q = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad \Omega = \Lambda^* Q \Lambda, \quad L = G - \Omega = G P_K. \quad (\text{HG6})$$

Keep $a_1 = b_N, a_2 = b_{N+1}, v = v_\lambda$, their original scale $\omega_N > 0, F_i = a_i^* G v, K_i = a_i^* L v, B_i = a_i^* \Omega v, E = v^* G v, d_i = a_i^* G a_i$ and $\kappa_N = \sqrt{d_1 d_2} / \omega_N$. ORE/FW [4] gives $\mathbf{c}_N = \overline{F_1 F_2} / (\omega_N E)$, with $U = K_1 / F_1, V = K_2 / F_2$. Substitution into the original products $\overline{U} V, (1 - \overline{U})(1 - V)$ and their complementary sum gives

$$\begin{aligned} \Phi_{K,N} &= 2\Re(\overline{K_1} K_2) / \omega_N, \\ \Phi_{B,N} &= 2\Re(\overline{B_1} B_2) / \omega_N, \\ \Phi_{\times,N} &= 2\Re(\overline{K_1} B_2 + \overline{B_1} K_2) / \omega_N. \end{aligned} \quad (\text{HG7})$$

This is an equality on the original nonzero-denominator domain and a polynomial continuation across vanishing approximate F_i . Both complementary cross products remain present.

Use the same vectors and ω_N for G' satisfying $(1 - \eta)G \preceq G' \preceq (1 + \eta)G, 0 < \eta \leq 1/4$. Taking the minimum on each unchanged observation fibre proves the same inequalities for Q , hence for Ω . Conjugation by $G^{-1/2}$ gives $|K'_i - K_i| \leq 2\eta\sqrt{d_i E}$ and $|B'_i - B_i| \leq \eta\sqrt{d_i E}$, exactly as HR3 [3]. Also $|K_i|, |B_i| \leq \sqrt{d_i E}$ because $0 \preceq L, \Omega \preceq G$. Expanding all linear and quadratic terms in (HG7) proves

$$\begin{aligned} |\tilde{\Phi}_K - \Phi_K| &\leq 2\kappa_N E(4\eta + 4\eta^2), \\ |\tilde{\Phi}_B - \Phi_B| &\leq 2\kappa_N E(2\eta + \eta^2), \\ |\tilde{\Phi}_\times - \Phi_\times| &\leq 2\kappa_N E(6\eta + 4\eta^2). \end{aligned} \quad (\text{HG8})$$

Since $|E' - E| \leq \eta E, E' \geq (1 - \eta)E$, and the three original absolute currents divided by $2E$ are at most $\kappa_N, \kappa_N, 2\kappa_N$, respectively,

$$\left| \frac{\tilde{\Phi}_{X,N}}{2E'} - \frac{\Phi_{X,N}}{2E} \right| \leq 12\kappa_N \eta, \quad X = K, B, \times. \quad (\text{HG9})$$

For example the cross bound is $\kappa_N(8\eta + 4\eta^2)/(1 - \eta) \leq 12\kappa_N \eta$; the other two are smaller. On the nonzero approximate denominator domain, HR's different approximation keeps $\mathbf{c}_N$ fixed. With $\mathbf{c}'_N = \overline{F'_1 F'_2} / (\omega_N E')$ its exact relation to (HG7) is

$$\tilde{\Phi}_{X,N} - \Phi_{X,N}^{\text{HR}} = 2E' \Re[(\mathbf{c}'_N - \mathbf{c}_N) P'_X]. \quad (\text{HG10})$$

Thus the two approximations are connected by a computed correction, and are not silently identified. In particular their complete sums retain this correction as well.

3 Whole-window heat depth and complete receiving error

On the simple quartet $q = (k+1)^2$ the existing FW48–50 estimates hold uniformly for $N = q-1+r, 0 \leq r \leq q+1$:

$$\begin{aligned} -\log p_{1,N} &= 2q\psi(r/q) + O_h(k + \log q), \quad |\log p_{2,N}| = O_h(\log q), \\ |U_N| &\leq e^{O_h(\varpi(k \log q))}, \quad |V_N| \leq C_h q/k, \quad \kappa_N^2 p_{1,N} p_{2,N} = |\mathbf{c}_N|^2. \end{aligned} \quad (\text{HG11})$$

The original signed correction in FW28 is $\mathbf{c}_N = k\delta + i(-1)^r(k\gamma - \varepsilon_N), 0 < \varepsilon_N = O_h(k^3/q^2)$, so $\log |\mathbf{c}_N| = O_h(\log q)$. The profile is strictly decreasing: FW44 gives $\psi'(t) = \log(\kappa_t/E(\kappa_t)) < 0$ for $t > 0$, and $\psi(0) = a_0 = \log(4/\pi)$. For an explicit check, use $r_t K/E = 1/(1+t)$ and $\psi(t) = \log(1+r_t) - (1+t)\log E + (t/2)\log(1-r_t^2)$. Differentiation using Carlson's formulas [9]

cancels the coefficient of r'_t exactly and leaves the stated derivative. Here $E \geq 1$ and $0 < \kappa_t < 1$ prove its sign. Therefore, with $p_*^{\text{win}} = \min_{N,i} p_{i,N}$,

$$-\log p_*^{\text{win}} = 2a_0q + O_h(k + \log q), \quad \log \max_N \kappa_N = a_0q + O_h(k + \log q). \quad (\text{HG12})$$

HM18–19 gives the relative metric comparison with $t_J = 2^{-(J+1)/2}c_*$ and $\eta_J = (1 + t_J)^{2k} - 1$, provided $t_J < 1$. For a specified $0 < \eta_* \leq 1/4$, a finite successful depth is

$$J = \max\{0, \lceil 2 \log_2(\bar{c}_k/d_*) \rceil - 1\}, \quad d_* = (1 + \eta_*)^{1/(2k)} - 1, \quad \bar{c}_k = C_h(2q + 3)B_h^{2q+2}. \quad (\text{HG13})$$

The least common strict HR denominator guard is $\eta_J < \sqrt{p_*^{\text{win}}}$. Inverting this increasing scalar function and retaining its strict integer rounding yields

$$J_{\text{guard}} + 1 = 2 \log_2 c_* + 2 \log_2(2k) - \log_2 p_*^{\text{win}} + O(1),$$

$$\lim_{k \rightarrow \infty} \frac{J_{\text{guard}}}{q} = \frac{2 \log(4/\pi)}{\log 2}. \quad (\text{HG14})$$

Indeed for $0 < x \leq 1$, $x/2 \leq \log(1+x) \leq x$ and $b \leq e^b - 1 \leq e^{1/2}b$ for $b = \log(1+x)/(2k)$. These give a bounded uniform error when taking logarithms of d_* . Integer rounding adds at most one. Insert (HG12) and $\log c_* = o(q)$ to get the limit.

For any fixed $\epsilon_0 > 0$, set $J_k = \lceil [2a_0/\log 2 + \epsilon_0]q \rceil$. Then $\eta_{J_k} = \exp[-a_0q - (\epsilon_0 \log 2/2)q + o_h(q)]$. HR6 gives $|U' - U| \leq \eta(2 + |U|)/(\sqrt{p_1} - \eta)$ and the identical formula for V . Substitution of (HG11)–(HG12), and $k \log q = o(q)$, proves

$$\max_N (|U_N^{[J_k]} - U_N| + |V_N^{[J_k]} - V_N|)$$

$$+ \sum_{N=q-1}^{2q-1} w_N \left| \frac{\tilde{\Phi}_{X,N}^{[J_k]}}{2E_N^{[J_k]}} - \frac{\Phi_{X,N}}{2E_N} \right| \leq e^{-(\epsilon_0 \log 2/2)q + o_h, \omega(q)}. \quad (\text{HG15})$$

The weights are the original $1, 2, \dots, 2, 1$, whose sum is $2q$. Their contribution has logarithm $O(\log q)$ and is absorbed in the displayed remainder. This controls approximation of each full signed current. It does not supply its unevaluated exact sign. For a determinant return of total absolute weighted rank B_{det} , the same minimum comparison instead gives $|\mathcal{L}' - \mathcal{L}| \leq B_{\text{det}}[-\log(1 - \eta)]$. The original polynomial-rank returns therefore require only $J = 2 \log_2 c_* + O(\log q + \log^+(1/\epsilon)) = o(q)$ at fixed tolerance.

4 The arithmetic action and its original boundary

Let $r = \text{rem}_h P$ and $\ell_h(P) = [w^{d-1}]r$. In the literal polynomial coordinate the heat source is $\mathcal{A}_N^{[J]}P = (I - T_s)P(D)a_h + T_sr(D)a_h$, $T_sf(x) = f(sx)$, $s = 2^{J+1}$. Euclidean division gives $wr - \text{rem}_h(wP) = h\ell_h(P)$. Since D commutes with T_s and $h(D)a_h = f_0 = \Theta\phi_*$,

$$D\mathcal{A}_N^{[J]}P - \mathcal{A}_{N+1}^{[J]}(wP) = T_sf_0\ell_h(P), \quad T_sf_0 = \Theta(T_s\phi_*). \quad (\text{HG16})$$

The original $\phi_* = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}$ is even, Schwartz, vanishes at zero and has zero integral. Dilation preserves these properties, so the displayed defect belongs to the same theta source. The fixed Hermite interpolation matrix sends $j_h P$ to $\ell_h(P)$. The proof of (HG5), with this fixed row in place of T_h , therefore gives $\log^+ \|\ell_h\|_{\mathcal{H}_N^*} = o_h(N)$. Here $\mathcal{H}_N$ is exactly the polynomial form in (HG1).

The original multinomial map is $P(S) \mapsto P(w_1 + \dots + w_k)$. Differentiate the tensor source in each physical variable and apply (HG16) in that factor. Its error is the sum of all k tensor terms, each with one factor $T_sf_0\ell_h$ and the remaining original heat source factors. HM's isometric original-source norm and $\|T_s\| = s^{-1/2}$ give

$$\|\mathfrak{D}_{k,N}^{[J]}\| \leq ks^{-1/2}\|f_0\|\|\ell_h\|_{\mathcal{H}_N^*}(1 + t_J)^{k-1}. \quad (\text{HG17})$$

Restricting through the multinomial map has norm one when its domain uses its pullback, the full original convolution form. Thus this bound keeps that norm, rather than an unrelated coefficient norm. At $J = \lceil \epsilon_0 q \rceil$ and $N \leq 2q + 2$ it is $\exp[-(\epsilon_0 \log 2/2)q + o_h(q)]$.

Let $\mathsf{T}_N : E_\chi \rightarrow \mathcal{P}_N$ be the original attained section for the full root polynomial, and $A = M_S$ on E_χ . Adding and subtracting the same polynomial lift proves exactly

$$\begin{aligned} D_{\text{tot}} \mathcal{A}_{k,N}^{[J]} \mathsf{T}_N - \mathcal{A}_{k,N+1}^{[J]} \mathsf{T}_{N+1} A &= \mathfrak{D}_{k,N}^{[J]} \mathsf{T}_N \\ &+ \mathcal{A}_{k,N+1}^{[J]} (M_S \mathsf{T}_N - \mathsf{T}_{N+1} A). \end{aligned} \quad (\text{HG18})$$

The parenthesized map has zero original remainder and takes values in $\chi \mathcal{P}_{N+1-q}$. Its entire image, and the heat term, are retained in the complete arithmetic action.

5 Back-propagation into the established native receivers

The full original observability theorem O1–12 [7] proves $\Lambda v_\lambda \neq 0$ on its explicit period locus. Consequently $e_{B,N} = (\Lambda v)^* Q_N \Lambda v = E_N (1 - \theta_N) > 0$ there. FW39 and NP19 [6] therefore apply without an undetected-class alternative. For heat constants $\alpha = (1 - t_J)^{2k}$, $\beta = (1 + t_J)^{2k}$, minimum comparison gives $\alpha/\beta \leq (1 - \theta'_N)/(1 - \theta_N) \leq \beta/\alpha$. The error in the original four-endpoint angle return is at most

$$4 \log(\beta/\alpha) = 8k \log \frac{1 + t_J}{1 - t_J}. \quad (\text{HG19})$$

There is no division by the possibly small observed fraction. Likewise the NP leakage Gram uses the same fixed action blocks in the original G_N ; its proved quantitative estimate stays

$$\sum_{j=0}^{r-1} (T^j)^* B^* Q_N B T^j \succeq (C_{R,N} C_{W,N} C_{T,N})^{-2} H_{K,N}. \quad (\text{HG20})$$

The new heat-growth theorem supplies finite approximation control for the metrics entering these original objects. It does not change the proper-source quotient into the different source-endpoint quotient, or discard the target minimum in NP30–31. The remaining native allocation and current values must still be evaluated in those maps.

6 A finite common guard with every fixed primary multiplicity

The sharp coefficient (HG14) uses the simple-packet profile. For the full packet $q = [1 + k(m_0 - 1)](k + 1)^2$, the incoming finite guard can also be proved without an observation inverse. Retain GRA3–6 [5]: put $p_h = 8m_0 - 9/2$, $B_h^{\text{tail}} = 42 + 8m_0$, $M_h^{\text{tail}} = 21 + 4m_0$, $\alpha_* = \pi/2$ and

$$\begin{aligned} \vartheta_h &= \int_{-1}^1 w_h(y) dy, \quad M_h(t) = \int_{\mathbb{R}} e^{ty} w_h(y) dy, \quad c_\Gamma = \sqrt{2} e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}} e^{2/3}, \\ \underline{a}_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha_*(k-3)} (k-2)^{-B_h^{\text{tail}}}}{C_\Gamma 2^{B_h^{\text{tail}}/2}}, \quad \bar{a}_k = \frac{C_h^{\text{tilt}}}{C_\Gamma} k^{p_h+1} M_h(\alpha_*)^{k-1}, \\ D_n &= [1 + 4(n + M_h^{\text{tail}})^2]^{M_h^{\text{tail}}}, \quad \frac{\underline{a}_k}{D_n} \|P\|_\sigma^2 \leq \|P\|_{w_h^{*k}}^2 \leq \bar{a}_k \|P\|_\sigma^2 \quad (\deg P \leq n). \end{aligned} \quad (\text{HG21})$$

Here c_h, C_h^{tilt} are precisely the full-source TW7 and BT12 constants retained in GRA4. The monic Gamma norm in this exact measure is $\gamma_n = \sqrt{2\pi}(2n)!/4^n$. For clarity GRA's polynomial comparison has no degree restriction: its recurrence $\|yP\|_\sigma \leq 2(n+1)\|P\|_\sigma$ gives $\int |P|^2 (1 + y^2)^{M_h^{\text{tail}}} d\sigma \leq D_n \|P\|_\sigma^2$. Cauchy–Schwarz and the complete density lower bound $w_h^{*k} \geq \underline{a}_k (1 + y^2)^{-M_h^{\text{tail}}}$ give the first inequality in (HG21); the full upper density gives the second. These retain all normalization masses.

Set $n_* = 2q + 2$, $R_k = k\sqrt{\delta^2 + \gamma^2}$ and $\tau = \pi/4$. For the complete monic root polynomial Q in $S = k/2 + iy$ all roots have modulus at most R_k , with every multiplicity retained. The literal monic trial $y^r Q(y)$ for the rank- r relation minimum ν_r obeys

$$\nu_r \leq \int (|y| + R_k)^{2(q+r)} w_h^{*k}(y) dy \leq 2e^{\tau R_k} M_h(\tau)^k (2q + 2r)! \tau^{-2q-2r}.$$

The second inequality uses $t^{2n} \leq (2n)! \tau^{-2n} e^{\tau t}$ and evenness of the full density. The original source minimum satisfies $\omega_n \geq \underline{a}_k \gamma_n / D_{n_*}$ for $n \leq n_*$. Therefore

$$\Delta_r = \omega_{q+r} / \nu_r \geq \underline{\Delta}_k, \quad \underline{\Delta}_k = \min \left\{ \frac{1}{2}, \frac{\underline{a}_k \sqrt{2\pi}}{2D_{n_*} e^{\tau R_k} M_h(\tau)^k} \left(\frac{\tau^2}{4} \right)^{n_*} \right\},$$

$$\bar{b}_k = \max\{1, (\bar{a}_k D_{n_*} / \underline{a}_k)(n_* + 1)^2\}, \quad \kappa_N \leq \bar{\kappa}_k = \sqrt{\bar{b}_k / \underline{\Delta}_k}, \quad p_{i,N} \geq \min\{1/2, k^2 \delta^2 / \bar{\kappa}_k^2\} =: p_*. \quad (\text{HG22})$$

Indeed (HG21) bounds ω_{n+1} / ω_n by $\bar{b}_k$. The original FW boundary-radius formula is $\kappa_{q-1}^2 = (\omega_q / \omega_{q-1})(\Delta_0^{-1} - 1)$ and, for $r = N + 1 - q \geq 1$, $\kappa_N^2 = (\omega_{N+1} / \omega_N)(1 - \Delta_{r-1})(\Delta_r^{-1} - 1)$. Every extra factor is at most its positive upper bound in (HG22). Finally $\kappa_N^2 p_{1,N} p_{2,N} = |\mathbf{c}_N|^2 \geq k^2 \delta^2$ and $p_{i,N} \leq 1$ prove both denominator bounds. GRA6 gives $\log(\bar{a}_k / \underline{a}_k) = O_h(k + \log q)$; each remaining logarithm in (HG22) is $O_h(q)$, hence $\log \bar{\kappa}_k + \log(1/p_*) = O_h(q)$.

For any prescribed $\varepsilon > 0$, choose

$$\eta_* \leq \min \left\{ \frac{1}{4}, \frac{\varepsilon}{32q\bar{\kappa}_k}, \frac{\sqrt{p_*}}{2}, \frac{\varepsilon p_*}{6}, \frac{\varepsilon}{2(B_{\det} + \varepsilon)} \right\}. \quad (\text{HG23})$$

Then (HG13) gives weighted current error at most ε , each response error at most ε by HR6–7, and determinant return error at most ε , using $-\log(1 - \eta_*) \leq 2\eta_*$. For the original polynomial-rank returns the resulting dyadic depth is $O_h(q + \log^+(1/\varepsilon))$. All finite guards are explicit; this is not a bound on the computational complexity of the integrals.

References

- [1] Supplied web continuation, *GitHub integration and further calculations*, 19 September 2026, equations (1)–(31), (63)–(64). Full mathematical response preserved in RECEIVED_MATHEMATICS.md.
- [2] *The circle heat source in the original arithmetic quotient metrics*, HM1–30; full provider included, public commit 607728ee3c35be98720d548794727670b1946c91.
- [3] *Heat approximation errors for the original complex observation responses*, HR1–11, same public edition; full provider included.
- [4] *Full-window source products and the retained class*, FW1–52, public commit 3c44de632b2c5ead13b23619d646803d2bbffc19; full provider included.
- [5] *Growing-degree original Gamma comparison*, GRA1–6, complete proof in the supplied cumulative signed-return source; original constants also in EC_COMPLETE.md, EC14–15.
- [6] *Native precision, marked detection and quantitative kernel leakage*, NP1–31, 19 September 2026; full preceding proof included.
- [7] *Full original observability*, O1–12, and *Original conductor coprimality*, NC1–27, public commit 12dd3645229bfebbf1cb8f5968ecffd4dfff1e9f.
- [8] T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, *Orthogonal Polynomials*, NIST DLMF Chapter 18, Table 18.9.1 and Section 18.3.

- [9] Bille C. Carlson, *Elliptic Integrals*, NIST DLMF Chapter 19, <https://dlmf.nist.gov/19.4>.
- [10] Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004, Appendix C.4.1, https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf.

Beurling Gram, complete source endpoint and arithmetic boundary

7 Objects, coordinates, and the exact original source

All scalar products are conjugate-linear in their first argument. Vectors are columns. A star means conjugate transpose; a bar on a matrix means entrywise conjugation. Fix the finite divisor from the original source,

$$h_Z(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \sum_{\rho \in Z} m_\rho, \quad 0 < \Re \rho < 1,$$

where each ρ is an actual nontrivial zero of ζ , and m_ρ is its full multiplicity. Each occurrence of a right divisor below means the restriction to the stipulated members with $\Re \rho > 1/2$. The statements preserve the divisor; they do not assert existence of an additional zero. The original jet coordinate is $u_{\rho,j} = [z^j]F(\rho + z)$, $0 \leq j < m_\rho$. Multiplication by s is

$$(A_Z u)_{\rho,j} = \rho u_{\rho,j} + u_{\rho,j-1}, \quad u_{\rho,-1} = 0. \quad (\text{EG1})$$

Thus $A_Z = \bigoplus_{\rho} (\rho I + J_{m_\rho}^\dagger)$. This lower triangular action is retained throughout.

Put $D = -x\partial_x$ and retain precisely

$$\begin{aligned} \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & \Theta f(x) &= 2 \sum_{n \geq 1} f(nx), \\ \Phi(s) &= \mathcal{M}\phi_*(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \\ g(s) &= 2\zeta(s)\Phi(s) = 2\xi(s). \end{aligned} \quad (\text{EG2})$$

The integral defining Φ converges for $\Re s > -2$. Its displayed formula follows by substituting $v = \pi x^2$ in the two Gaussian integrals and using $\Gamma(z+1) = z\Gamma(z)$ twice. In particular $\Phi(\rho) \neq 0$. The polynomial $r_Z(u)$ is the unique polynomial of degree $< d$ whose full jets are $j_Z(h_Z/g)u$. This polynomial exists uniquely by the finite Chinese remainder theorem: the factors $(s - \rho)^{m_\rho}$ are pairwise coprime, and reduction of polynomials of degree $< d$ is an isomorphism onto their product. This retains the complete local unit h_Z/g , not merely its constant term.

Write its proper rational function as

$$\frac{r_Z(u)(s)}{h_Z(s)} = \sum_{\rho \in Z} \sum_{\ell=1}^{m_\rho} \frac{v_{\rho,\ell}(u)}{(s - \rho)^\ell}. \quad (\text{EG3})$$

The exact reconstructed source is

$$\psi_u(x) = \sum_{\rho, \ell} v_{\rho,\ell}(u) \frac{x^{-\rho}}{(\ell-1)!} \int_x^\infty \phi_*(y) y^{\rho-1} (\log(y/x))^{\ell-1} dy. \quad (\text{EG4})$$

This is the original MI6 source, with every primary and the original $1/2$ in (EG2). Indeed, for $\Re s > 1$, absolute integration and the substitution $x = ye^{-t}$ give

$$\int_0^y x^{s-\rho-1} \frac{(\log(y/x))^{\ell-1}}{(\ell-1)!} dx = \frac{y^{s-\rho}}{(s-\rho)^\ell}.$$

Consequently

$$\mathcal{M}\psi_u(s) = \Phi(s) \frac{r_Z(u)(s)}{h_Z(s)}, \quad \mathcal{M}\Theta\psi_u(s) = g(s) \frac{r_Z(u)(s)}{h_Z(s)} = \mathcal{M}R_Z u(s). \quad (\text{EG5})$$

All Gaussian tails are absolutely summable, including their divisor weighted sums, so Mellin uniqueness and the finite divisor identity $\sum_{n|k} \mu(n) = \mathbf{1}_{k=1}$ also give $\psi_u = \frac{1}{2} \sum_{n \geq 1} \mu(n) (R_Z u)(n \cdot)$. Thus (EG4) is an exact formula for the original reconstruction. Fourier/Mellin uniqueness and the retained $1/(2\pi)$ convention are as in the transform formulas of [2]. The original zero-integral source condition is preserved as well: each singular term is integrable at zero because $\Re \rho < 1$, and the source decays at infinity. Evaluating its legitimate Mellin integral at $s = 1$ gives $\int_0^\infty \psi_u(x) dx = \Phi(1) r_Z(u)(1)/h_Z(1) = 0$.

Proposition 1 (The entire principal-part map). *Write $\zeta(\rho + z) = z^{m_\rho} \zeta_\rho(z)$, with $\zeta_\rho(0) = \zeta^{(m_\rho)}(\rho)/m_\rho! \neq 0$. Define*

$$a_{\rho,j} = [z^{m_\rho-j-1}] \frac{\sum_{l=0}^{m_\rho-1} u_{\rho,l} z^l}{2\zeta_\rho(z)}, \quad a = U_Z u. \quad (\text{EG6})$$

Then U_Z is invertible, and its exact block determinant is

$$\det U_\rho = (-1)^{m_\rho(m_\rho-1)/2} (2\zeta_\rho(0))^{-m_\rho}. \quad (\text{EG7})$$

Put $A_{\text{pp}} = \bigoplus_\rho (\rho I + J_{m_\rho}^\dagger)$. The exact action equation is

$$U_Z A_Z = A_{\text{pp}} U_Z. \quad (\text{EG8})$$

For $0 < x < 1$, the source is

$$\psi_u(x) = \sum_{\rho,j} a_{\rho,j} x^{-\rho} \frac{(-\log x)^j}{j!} + r_u(x), \quad r_u(x) = O_Z(\|u\|x^2). \quad (\text{EG9})$$

The remainder estimate holds for every fixed number of D -derivatives.

Proof. The full jet of $\mathcal{M}R_Z u$ at ρ is u_ρ by its defining unit. Division by 2ζ in (EG5) therefore gives exactly (EG6). Multiplication by the Taylor series of $(2\zeta_\rho)^{-1}$ is lower triangular with diagonal $(2\zeta_\rho(0))^{-1}$. Reversing its m_ρ output coefficients has determinant $(-1)^{m_\rho(m_\rho-1)/2}$, proving (EG7). An explicit inverse is

$$u_{\rho,l} = [z^l] \left(2\zeta_\rho(z) \sum_{j=0}^{m_\rho-1} a_{\rho,j} z^{m_\rho-j-1} \right).$$

Multiplication of $\sum_j a_{\rho,j} z^{-j-1}$ by $s = \rho + z$ changes its principal coefficient of z^{-j-1} to $\rho a_{\rho,j} + a_{\rho,j+1}$, where $a_{\rho,m_\rho} = 0$. The discarded constant is holomorphic and has no principal part. Since multiplication commutes with the local zeta unit, this proves (EG8) on the original jet space.

For each summand in (EG4), replace its lower integration limit by zero. Expanding $(\log y - \log x)^{\ell-1}$ gives

$$\frac{x^{-\rho}}{(\ell-1)!} \sum_{q=0}^{\ell-1} \binom{\ell-1}{q} (-\log x)^{\ell-1-q} \Phi^{(q)}(\rho).$$

These are exactly the Laurent coefficients just calculated. The difference is minus the integral from zero to x . Substitution $y = xv$ shows it is $O(x^2)$, since $\phi_*(xv) = O(x^2 v^2)$ and $\Re \rho + 2 > 0$. More explicitly, for $\phi_*(x) = \sum_{n \geq 1} c_{2n} x^{2n}$ it is the convergent series

$$(-1)^\ell \sum_{n \geq 1} \frac{c_{2n}}{(\rho + 2n)^\ell} x^{2n}.$$

Termwise D -differentiation proves the derivative assertion. The Gaussian integral from x to infinity proves rapid decay of (EG4) at infinity. These facts prove every endpoint assertion. $\square$

8 The Beurling Gram, its inverse, and its displacement

In this section $Z = Z_+$ is a right divisor. Retain

$$H_B = L^2((1, \infty), y^{-2} dy), \quad (J_+ f)_{\rho, j} = \frac{1}{j!} \int_1^\infty f(y) (-\log y)^j y^{-\rho-1} dy.$$

The representers are $k_{\rho, j}(y) = (-\log y)^j y^{1-\bar{\rho}}/j!$. Changing variables $t = \log y$ gives

$$\mathbf{G}_{(\rho, j), (\sigma, l)} = \langle k_{\rho, j}, k_{\sigma, l} \rangle = \frac{(-1)^{j+l} \binom{j+l}{j}}{(\rho + \bar{\sigma} - 1)^{j+l+1}}, \quad b_{\rho, j} = (J_+ 1)_{\rho, j} = \frac{(-1)^j}{\rho^{j+1}}. \quad (\text{EG10})$$

The integration identity used here follows by integrating the exponential $j + l$ times by parts; its boundary terms vanish because $\Re(\rho + \bar{\sigma} - 1) > 0$. The exponential polynomials defining the representers are independent. To prove this, fix one exponent and apply all the other factors $(\partial_t - (1 - \bar{\sigma}))^{m_\sigma}$ to a putative relation. They kill the other exponents and act invertibly on the fixed polynomial factor: each such factor is a nonzero scalar plus a nilpotent derivative. Thus its polynomial must be zero. Repeating for each exponent proves independence. Hence $\mathbf{G} = J_+ J_+^* > 0$.

Theorem 1 (Complete finite Beurling projection). *Let*

$$B_Z(s) = \prod_\rho \left(\frac{s - \rho}{s + \bar{\rho} - 1} \right)^{m_\rho}, \quad c_Z = \overline{B_Z(1)}, \quad S_Z(s) = \frac{1 - c_Z B_Z(s)}{s}, \quad R_Z^B(s) = \frac{c_Z B_Z(s)}{s}. \quad (\text{EG11})$$

Then the orthogonal decomposition of 1 into $\text{ran } J_+^$ and $\ker J_+$ has respective transforms S_Z and R_Z^B , and*

$$b^* \mathbf{G}^{-1} b = 1 - \prod_{\rho \in Z} \left| \frac{1 - \rho}{\rho} \right|^{2m_\rho}. \quad (\text{EG12})$$

For the original Beurling source space $V_Q = \text{span}\{\alpha[y] - [\alpha y] : \alpha = a/Q, 1 \leq a < Q\}$, writing $1 = s_Z + r_Z$ for that orthogonal decomposition, the full error is exactly

$$\text{dist}(1, V_Q)^2 = b^* \mathbf{G}^{-1} b + \text{dist}(r_Z, V_Q)^2. \quad (\text{EG13})$$

Proof. Use the transform $\mathcal{L}_B f(s) = \int_1^\infty f(y) y^{-s-1} dy$. For the representer it is $(-1)^j / (s + \bar{\rho} - 1)^{j+1}$. For each factor of B_Z , direct substitution proves $\overline{B_Z(1)} B_Z(0) = 1$; on $\Re s = 1/2$, numerator and denominator have equal modulus. Thus S_Z has no pole at zero and has only the displayed left-half-plane poles. It is proper at infinity, so its partial fractions are the transforms of a unique linear combination of the $k_{\rho, j}$. The coefficients of its highest poles are nonzero because $0 < \Re \rho < 1$ and the right and left divisors are disjoint. Hence the full multiplicities remain. Define s_Z by these partial fractions and $r_Z = 1 - s_Z$. Its transform is R_Z^B , which vanishes at every ρ to the full order m_ρ . Consequently $J_+ r_Z = 0$, proving orthogonality.

The map $f \mapsto e^{-t/2} f(e^t)$ is an isometry $H_B \rightarrow L^2(0, \infty; dt)$. On $s = 1/2 + i\tau$, its Fourier transform is precisely $\mathcal{L}_B f(s)$. Plancherel, in this convention, gives

$$\|r_Z\|^2 = \frac{|c_Z|^2}{2\pi} \int_{\mathbb{R}} \frac{d\tau}{1/4 + \tau^2} = |c_Z|^2.$$

The integral is 2π by the antiderivative $2 \arctan(2\tau)$. Since $\|1\|^2 = 1$, this proves (EG12). For $F_\alpha = \alpha[y] - [\alpha y]$, integrating the two floor-function step sums first on $\Re s > 1$ gives $\mathcal{L}_B F_\alpha(s) = \zeta(s)(\alpha - \alpha^s)/s$. The boundedness of F_α continues the identity to $\Re s > 0$; the pole at 1 is removable. Differentiating through all retained orders gives $J_+ F_\alpha = 0$. Thus $V_Q \subset \ker J_+$, and Pythagoras followed by the infimum proves (EG13). The finite-primary projection has the classical context discussed by Burnol [4]; its exact value here follows from the displayed calculation, without using a complete approximation theorem. $\square$

For completeness every inverse coefficient is explicit. If $a = G^{-1}b$, then

$$a_{\rho,j} = \frac{(-1)^j}{(m_\rho - j - 1)!} \left[\frac{d^{m_\rho-j-1}}{ds^{m_\rho-j-1}} ((s + \bar{\rho} - 1)^{m_\rho} S_Z(s)) \right]_{s=1-\bar{\rho}}. \quad (\text{EG14})$$

Here a is a Beurling representer coefficient, distinguished by its stated domain from the principal-part coefficient in (EG6). For arbitrary jet data u , put $H(s) = \prod(s-\rho)^{m_\rho}$, $D_B(s) = \prod(s+\bar{\rho}-1)^{m_\rho}$, $H_\rho = H/(s-\rho)^{m_\rho}$, and $U_\rho(s) = \sum_{j < m_\rho} u_{\rho,j}(s-\rho)^j$. Then

$$P_u(s) = \sum_\rho H_\rho(s) \text{rem}_{(s-\rho)^{m_\rho}} \frac{D_B(s)U_\rho(s)}{H_\rho(s)}. \quad (\text{EG15})$$

The remainder means the finite Taylor polynomial of the displayed holomorphic germ. Each summand has degree at most $d-1$. Reducing at a fixed root kills all the other summands and gives $D_B U_\rho$. Since D_B is a unit there, P_u/D_B has exactly the prescribed jets. Its proper partial fractions, by (EG14) with S_Z replaced by P_u/D_B , give $G^{-1}u$. This proves both its existence and uniqueness without omitting a derivative slot.

Theorem 2 (Confluent determinant and the exact action defect). *For any fixed ordering of the distinct roots,*

$$\det G = \frac{\prod_{\rho < \sigma} |\rho - \sigma|^{2m_\rho m_\sigma}}{\prod_\rho (2\Re \rho - 1)^{m_\rho^2} \prod_{\rho < \sigma} |\rho + \bar{\sigma} - 1|^{2m_\rho m_\sigma}}. \quad (\text{EG16})$$

Let $e_{\rho,j} = \mathbf{1}_{j=0}$ and $H_B^Z = G^{-1}$. Then

$$A_Z G + G A_Z^* - G = ee^*, \quad (\text{EG17})$$

$$A_Z^* H_B^Z + H_B^Z A_Z - H_B^Z = H_B^Z ee^* H_B^Z, \quad (\text{EG18})$$

$$e^* G^{-1} e = \sum_\rho m_\rho (2\Re \rho - 1). \quad (\text{EG19})$$

The relative defect of H_B^Z has exactly this one positive eigenvalue and $d-1$ zero eigenvalues.

Proof. The ordinary Cauchy formula is

$$\det \left[\frac{1}{x_i + \bar{x}_j - 1} \right] = \frac{\prod_{i < j} |x_i - x_j|^2}{\prod_{i,j} (x_i + \bar{x}_j - 1)}.$$

It is the sign-preserving specialization of [1, (1.3.14)]. It also follows directly by multiplying by its denominator: the result is alternating separately in the row and column variables, has exactly the degrees of the two Vandermonde products, and its constant is one by successive residues at a matched row and column pole (the one-by-one determinant starts the induction). Replace the m_ρ copies of a root by distinct points tending to ρ . Within each cluster apply Newton divided differences to rows and columns. Their determinant divides by the corresponding Vandermonde. A divided difference of order j tends to the j th derivative divided by $j!$, as follows from Taylor expansion or the finite interpolation formula. The limiting entry is $(-1)^{j+l} (j+l)! / [j!l!(\rho + \bar{\sigma} - 1)^{j+l+1}]$. All within-cluster Vandermondes have been divided out. The cross-cluster factors have limits as in (EG16); the within-cluster denominator is $(2\Re \rho - 1)^{m_\rho^2}$. This proves the formula and the absence of any additional factorial.

Let $z = \rho + \bar{\sigma} - 1$. The $(\rho, j), (\sigma, l)$ entry on the left of (EG17) is $z G_{j,l} + \mathbf{1}_{j>0} G_{j-1,l} + \mathbf{1}_{l>0} G_{j,l-1}$. For $j+l > 0$ its coefficient is $(-1)^{j+l} \{ \binom{j+l}{j} - \binom{j+l-1}{j-1} - \binom{j+l-1}{j} \} = 0$, where absent binomial terms mean zero. For $j=l=0$ it is 1. Thus (EG17) holds. Multiplication by G^{-1} on both sides gives (EG18). Multiplying only on one side and taking trace gives $e^* G^{-1} e = 2\Re \text{Tr } A_Z - d$, proving (EG19). Finally the relative Hermitian matrix is $(H_B^Z)^{1/2} ee^* (H_B^Z)^{1/2}$, whose only nonzero eigenvalue is the squared norm of $(H_B^Z)^{1/2} e$. This proves the spectral claim. $\square$

9 Explicit source remainder bounds

For a principal-part vector a , set

$$\Psi_{\text{pp}}(x)a = \psi_{\mathcal{U}_Z^{-1}a}(x), \quad p(x)a = \mathbf{1}_{(0,1)}(x) \sum_{\rho,j} a_{\rho,j} x^{-\rho} \frac{(-\log x)^j}{j!}, \quad r(x) = \Psi_{\text{pp}}(x) - p(x). \quad (\text{EG20})$$

These are actual row-valued functions on the original x coordinate. Let W be the finite map from a to the coefficients v in (EG3). Explicitly,

$$(Wa)_{\rho,\ell} = [z^{m_\rho-\ell}] \frac{\sum_{j=0}^{m_\rho-1} a_{\rho,j} z^{m_\rho-j-1}}{\Phi(\rho+z)}. \quad (\text{EG21})$$

Indeed multiplication of the principal part of $\mathcal{M}\psi$ by the analytic unit $1/\Phi$ recovers the proper rational function's principal parts. The difference has no pole and is proper, hence is zero. This proves (EG21), including all of its local derivatives.

Lemma 1 (A fully finite bound from the MI6 integrals). *Put $C_\phi = 4\pi^2 + 6\pi$ and $\beta_\rho = \Re \rho$. For one MI6 summand of order ℓ , its remainder defined as in (EG20) has squared $L^2(dx)$ norm at most*

$$B_{\rho,\ell}^2 = \frac{C_\phi^2}{5(\beta_\rho + 2)^{2\ell}} + \frac{1}{2} \left[\frac{C_\phi}{2(\ell-1)!} (\pi/2)^{-(\beta_\rho+\ell+3)/2} \Gamma((\beta_\rho + \ell + 3)/2) \right]^2. \quad (\text{EG22})$$

Consequently the explicit constants

$$R_0 = \|W\| \left(\sum_{\rho,\ell} B_{\rho,\ell}^2 \right)^{1/2}, \quad K_r = C_\phi \|W\| \left(\sum_{\rho,\ell} (\beta_\rho + 2)^{-2\ell} \right)^{1/2} \quad (\text{EG23})$$

satisfy $\|a \mapsto r(\cdot)a\| \leq R_0$ and $\|r(x)\| \leq K_r x^2$ for $0 < x < 1$.

Proof. For $0 < y < 1$, $|\phi_*(y)| \leq C_\phi y^2$. The integral from zero to x for the regular part of one summand is therefore bounded by

$$\frac{C_\phi x^{-\beta}}{(\ell-1)!} \int_0^x y^{\beta+1} |\log(y/x)|^{\ell-1} dy = \frac{C_\phi x^2}{(\beta+2)^\ell}.$$

Its squared integral is the first term of (EG22). For $y \geq x \geq 1$, $|\phi_*(y)| \leq C_\phi y^4 e^{-\pi y^2}$, $\log(y/x) \leq y$, and $e^{-\pi y^2} \leq e^{-\pi x^2/2} e^{-\pi y^2/2}$. The tail integral is thus bounded by

$$\frac{C_\phi}{2(\ell-1)!} (\pi/2)^{-(\beta+\ell+3)/2} \Gamma((\beta+\ell+3)/2) x^{-\beta} e^{-\pi x^2/2}.$$

Because $\beta > 0$, $\int_1^\infty x^{-2\beta} e^{-\pi x^2} dx \leq \int_1^\infty e^{-2x} dx < 1/2$. This proves the second term. For a linear combination, Cauchy–Schwarz over the finite vector of coefficients Wa gives respectively the two bounds in (EG23). No growing matrix is inverted in these constants. $\square$

10 Exact right-endpoint metric and its finite correction

Define the actual cutoff metric by

$$u^* \Gamma_Z(T) u = \int_{e^{-T}}^\infty |\psi_u(x)|^2 dx, \quad T \geq 0, \quad \Gamma_{\text{pp}}(T) = \mathcal{U}_Z^{-*} \Gamma_Z(T) \mathcal{U}_Z^{-1}. \quad (\text{EG24})$$

It is positive definite at every finite T . Indeed, a zero norm means the finite analytic sum (EG4) vanishes on an interval, and hence everywhere on $(0, \infty)$. Its principal coefficients and then u vanish by Proposition 1.

For a right divisor put

$$\lambda_\rho = \rho - \frac{1}{2}, \quad d_\rho = \Re \lambda_\rho > 0, \quad L = A_{\text{pp}} - \frac{1}{2}I, \quad E_T = e^{TL} = \bigoplus_{\rho} e^{\lambda_\rho T} e^{T J_{m_\rho}^\dagger}, \quad \widehat{G} = \overline{G}. \quad (\text{EG25})$$

The reversed row is

$$q(\tau)_{\rho,j} = e^{-\lambda_\rho \tau} \frac{(-\tau)^j}{j!} = (e^* e^{-\tau L})_{\rho,j}. \quad (\text{EG26})$$

The original logarithmic singular row is $f(t) = e^* e^{tL}$. Thus $f(T - \tau) = q(\tau) E_T$, including the entire nilpotent translation. Integrating gives $\int_0^\infty q(\tau)^* q(\tau) d\tau = \widehat{G}$.

Theorem 3 (Exact finite correction and improved endpoint rate). *The singular cutoff Gram satisfies the exact identity*

$$S(T) := \int_0^T f(t)^* f(t) dt = E_T^* \widehat{G} E_T - \widehat{G}. \quad (\text{EG27})$$

Both source cross terms and the entire regular norm converge in the explicit Hermitian matrix

$$F_Z = \int_0^\infty \{p(x)^* r(x) + r(x)^* p(x) + r(x)^* r(x)\} dx, \quad K_Z = F_Z - \widehat{G}. \quad (\text{EG28})$$

Define, with the same integrand,

$$B_Z^{\text{reg}}(T) = \int_0^{e^{-T}} \{p^* r + r^* p + r^* r\} dx. \quad (\text{EG29})$$

Then the exact source metric identity is

$$\boxed{\Gamma_{\text{pp}}(T) = E_T^* \widehat{G} E_T + K_Z - B_Z^{\text{reg}}(T)}. \quad (\text{EG30})$$

In particular, if $d_* = \min d_\rho$, $m_* = \max m_\rho$, and $P_*(T) = \sum_{j=0}^{m_*-1} T^j / j!$, then

$$\|E_T^{-*} \Gamma_{\text{pp}}(T) E_T^{-1} - \widehat{G}\| \leq e^{-2d_* T} P_*(T)^2 \{K_0 + \mathcal{B}_Z(T)\}, \quad (\text{EG31})$$

where every constant and function on the right is finite and explicit:

$$K_0 = \|G\| + R_0^2 + 2K_r \sum_{\rho,j} (3 - \beta_\rho)^{-j-1}, \quad (\text{EG32})$$

$$\mathcal{B}_Z(T) = 2K_r \sum_{\rho,j} e^{-(3-\beta_\rho)T} \sum_{k=0}^j \frac{T^{j-k}}{(j-k)!(3-\beta_\rho)^{k+1}} + \frac{K_r^2}{5} e^{-5T}. \quad (\text{EG33})$$

The summation convention here and below is $\rho \in Z$ and $0 \leq j < m_\rho$.

Proof. Conjugating (EG17) gives $L^* \widehat{G} + \widehat{G} L = e e^*$, because $A_{\text{pp}} = A_Z^T$. Differentiating $E_T^* \widehat{G} E_T$ therefore gives $f(T)^* f(T)$; at $T = 0$ its value is $\widehat{G}$. This proves (EG27). Equivalently, the omitted reversed tail transforms exactly to $\widehat{G}$ because $q(\tau + T) E_T = q(\tau)$.

The cross integrals converge absolutely. On $(0, 1)$ the bounds are

$$\|r(x)\| \leq K_r x^2, \quad \|p(x)\| \leq \sum_{\rho,j} x^{-\beta_\rho} (-\log x)^j / j!.$$

On $[1, \infty)$, $p = 0$ and r is square integrable. Also

$$\int_0^1 x^{2-\beta} \frac{(-\log x)^j}{j!} dx = (3 - \beta)^{-j-1}.$$

It follows that $\|K_Z\| \leq K_0$. Expanding $(p+r)^*(p+r)$ in the original integral gives (EG30), with neither cross term removed. For the small-interval tail, repeated integration by parts gives

$$\int_0^{e^{-T}} x^{2-\beta} \frac{(-\log x)^j}{j!} dx = e^{-(3-\beta)T} \sum_{k=0}^j \frac{T^{j-k}}{(j-k)!(3-\beta)^{k+1}}.$$

Together with $\int_0^{e^{-T}} K_r^2 x^4 dx = K_r^2 e^{-5T}/5$, this proves $\|B_Z^{\text{reg}}(T)\| \leq \mathcal{B}_Z(T)$. Finally $\|(J_m^\dagger)^j\| \leq 1$ and the nilpotent exponential is finite, so $\|\mathbf{E}_T^{-1}\| \leq e^{-d_* T} P_*(T)$. Congruencing (EG30) by $\mathbf{E}_T^{-1}$ proves (EG31). $\square$

The originally proposed error bound also follows, with its exact constants. The positive tail Gram of q has norm at most its trace

$$\mathcal{T}_Z(T) = \sum_{\rho, j} \frac{(2j)! e^{-2d_\rho T}}{(j!)^2 (2d_\rho)^{2j+1}} \sum_{l=0}^{2j} \frac{(2d_\rho T)^l}{l!}. \quad (\text{EG34})$$

Indeed its diagonal entries are $\int_T^\infty e^{-2d_\rho t} t^{2j} / (j!)^2 dt$, evaluated by repeated integration by parts. The reversed singular map has norm at most $\sqrt{\|\mathbf{G}\|}$, while the transformed regular map has norm at most $R_0 e^{-d_* T} P_*(T)$. Expanding the Gram of their sum proves the original upper bound

$$\mathcal{T}_Z(T) + 2\sqrt{\|\mathbf{G}\|} R_0 e^{-d_* T} P_*(T) + R_0^2 e^{-2d_* T} P_*(T)^2. \quad (\text{EG35})$$

Thus the supplied (48)–(51) are valid. The stronger (EG31) uses the actual $O(x^2)$ remainder to keep the finite cross integrals, instead of estimating them only by Cauchy–Schwarz against the growing singular norm.

An explicit relative lower bound is

$$g_* = \frac{\det \mathbf{G}}{(\text{Tr } \mathbf{G})^{d-1}} \leq \lambda_{\min}(\mathbf{G}). \quad (\text{EG36})$$

If the positive eigenvalues are $\lambda_1, \dots, \lambda_d$, the product of the $d-1$ factors other than a smallest one is at most $(\sum \lambda_i)^{d-1}$, proving the inequality (also when $d=1$). Division of either error bound by g_* is therefore a fully specified relative metric bound.

Theorem 4 (Determinant, including the finite term). *For every right divisor,*

$$\begin{aligned} \log \det \Gamma_Z(T) &= 2T \sum_{\rho} m_{\rho} (\Re \rho - \tfrac{1}{2}) - 2 \sum_{\rho} m_{\rho} \log \left(2 \left| \frac{\zeta^{(m_{\rho})}(\rho)}{m_{\rho}!} \right| \right) + \log \det \mathbf{G} \\ &\quad + O_Z \left(e^{-2d_* T} (1+T)^{2m_*-2} \right). \end{aligned} \quad (\text{EG37})$$

No $\log T$ term is present. For a right conjugate pair $\rho = 1/2 + \delta + i\gamma$, $\bar{\rho}$, with $\delta > 0$, $\gamma \neq 0$ and common full multiplicity m , its constant term is explicitly

$$\begin{aligned} \log \det \Gamma_+(T) &= 4m\delta T - 4m \log \left(2 \left| \frac{\zeta^{(m)}(\rho)}{m!} \right| \right) \\ &\quad + 2m^2 \log \frac{|\gamma|}{2\delta\sqrt{\delta^2 + \gamma^2}} + O_Z(e^{-2\delta T} (1+T)^{2m-2}). \end{aligned} \quad (\text{EG38})$$

On the stipulated original convention $\gamma > 0$, this is exactly the constant in the supplied (53).

Proof. The nilpotent exponential is upper triangular with diagonal one, so $|\det \mathbf{E}_T|^2 = \exp(2T \sum m_{\rho} d_{\rho})$. Proposition 1 gives the exact $|\det \mathbf{U}_Z|^2$. Let $X_T = \widehat{\mathbf{G}}^{-1/2} \mathbf{E}_T^*(K_Z -$

$B_Z^{\text{reg}}(T))E_T^{-1}\widehat{G}^{-1/2}$. Its norm is at most (EG31) divided by g_* . For all sufficiently large T this is less than one, and

$$|\log \det(I + X_T)| \leq d \frac{\|X_T\|}{1 - \|X_T\|}$$

by applying $|\log(1 + x)| \leq |x|/(1 - |x|)$ to each real eigenvalue. Taking determinants in (EG30) proves (EG37). A still more precise finite correction is

$$\log \det(I + X_T) = \text{Tr } X_T + \mathcal{R}_T, \quad |\mathcal{R}_T| \leq \frac{d\|X_T\|^2}{2(1 - \|X_T\|)}.$$

This follows from the convergent power series of $\log(1 + x)$; X_T retains the exact K_Z and the exact small-interval tail. For the pair, (EG16) gives

$$\det G = \frac{(2|\gamma|)^{2m^2}}{(2\delta)^{2m^2}(2\sqrt{\delta^2 + \gamma^2})^{2m^2}}.$$

Zeta commutes with conjugation, so the two absolute local units are equal. Substitution proves (EG38). $\square$

11 Keeping the entire left block before the minimum

The same finite source (EG4) respects extension by zero from a subset of the divisor: multiplying its polynomial r_Z and denominator h_Z by the same extra factors preserves their quotient. Thus the following block decomposition is one of the actual original section, not a replacement section.

Proposition 2 (Exact finite quartet minimum). *Split a full quartet, or any finite divisor without a critical member, as $Z_+ \sqcup Z_-$. The left Gram converges to a positive definite matrix $\Gamma_{--}(\infty)$ in the original dx norm. The attained right-value quotient is exactly*

$$Q_+(T) = \Gamma_{++}(T) - \Gamma_{+-}(T)\Gamma_{--}(T)^{-1}\Gamma_{-+}(T). \quad (\text{EG39})$$

With the original right maps $U_+, E_{+,T}$,

$$E_{+,T}^{-*}U_+^{-*}Q_+(T)U_+^{-1}E_{+,T}^{-1} \longrightarrow \widehat{G}_+. \quad (\text{EG40})$$

Consequently (EG38), with its error term replaced by $o(1)$, also gives the quotient determinant for the original quartet.

Proof. For a left member $\beta < 1/2$, each singular function $x^{-\rho}(-\log x)^j$ is in $L^2(0, 1; dx)$. The remainder is also in L^2 , so the left Gram converges entrywise. A vector in the kernel of its limit would give a source identically zero; its exact principal coefficients force the vector to be zero. Thus its limit is positive definite. At finite T , completing the square in the actual block quadratic form gives (EG39), with minimizer $u_- = -\Gamma_{--}(T)^{-1}\Gamma_{-+}(T)u_+$.

Here is the required cross-block convergence. In $t = -\log x$, the transformed right singular row on $0 \leq t \leq T$ is $q_+(T - t)$, whose norm is uniformly bounded in $L^2(0, T)$. For $0 \leq t \leq T/2$ its squared norm is bounded by the tail (EG34) at $T/2$, and hence tends to zero. On $T/2 \leq t \leq T$ the L^2 norm of any fixed left source tends to zero, since the left source is square integrable on the full logarithmic line. Cauchy–Schwarz on the two intervals shows its inner product with the transformed right singular source tends to zero. The transformed right regular source has full norm at most $R_0\|E_{+,T}^{-1}\|$, also tending to zero; this also covers $t < 0$, where the singular cutoff row is zero. Since the left domain is finite dimensional, the transformed cross-block converges to zero in operator norm. The left inverses remain bounded, so their complete Schur correction tends to zero. Theorem 3 proves (EG40). Taking determinants, with the same $U_+, E_{+,T}$, gives the claimed constant. This quotient is precisely E_Z/E_{Z_-} with value $[u_+ + u_-] \mapsto u_+$; no other programme quotient is used. $\square$

Proposition 3 (A finite bound after the original quartet minimum). *Retain the original quartet $1/2 \pm \delta \pm i\gamma$, with $0 < \delta < 1/2$, $\gamma \neq 0$, and full common multiplicity m . Use principal-part coordinates separately on both sides. Write $\Gamma_{\text{pp},--}(T) = \mathbf{U}_{-}^{-*} \Gamma_{--}(T) \mathbf{U}_{-}^{-1}$, and set*

$$g_{-} = \frac{\det \Gamma_{\text{pp},--}(1)}{(\text{Tr } \Gamma_{\text{pp},--}(1))^{2m-1}} > 0. \quad (\text{EG39a})$$

This is a fixed matrix of the explicit MI6 integrals. Let $R_{0,\pm}$, $K_{r,\pm}$ be the constants (EG23) for the respective original restricted source maps, and put

$$C_{+-} = K_{r,-} \sum_{\rho \in Z_{+,j}} (3 - \beta_{\rho})^{-j-1} + K_{r,+} \sum_{\sigma \in Z_{-,l}} (3 - \beta_{\sigma})^{-l-1} + R_{0,+} R_{0,-},$$

$$P_m(T) = \sum_{j=0}^{m-1} \frac{T^j}{j!}, \quad V_m(T) = \sum_{j=0}^{m-1} \sum_{l=0}^{m-1} \frac{T^{j+l+1}}{(j+l+1)!}, \quad \eta_{+-}(T) = e^{-\delta T} \{2V_m(T) + C_{+-}P_m(T)\}. \quad (\text{EG39b})$$

Then for $T \geq 1$ the transformed Schur correction has norm at most $\eta_{+-}(T)^2/g_{-}$. In particular, with $\varepsilon_{+}(T)$ the right side of (EG31) for Z_{+} ,

$$\left\| \mathbf{E}_{+,T}^{-*} \mathbf{U}_{+}^{-*} Q_{+}(T) \mathbf{U}_{+}^{-1} \mathbf{E}_{+,T}^{-1} - \widehat{\mathbf{G}}_{+} \right\| \leq \varepsilon_{+}(T) + \frac{\eta_{+-}(T)^2}{g_{-}}. \quad (\text{EG39c})$$

Consequently the right-value quotient has the full determinant constant in (EG38), with the proved remainder $O_Z(e^{-2\delta T}(1+T)^{4m-2})$.

Proof. An invertible block diagonal congruence by $\mathbf{U}_{+}^{-1} \oplus \mathbf{U}_{-}^{-1}$ carries the original block metric and its attained Schur complement to their principal-part versions. It preserves the minimizing operation: the original minimizing u_{-} corresponds bijectively to $a_{-} = \mathbf{U}_{-} u_{-}$. Thus it is enough to bound the transformed principal-part cross-block

$$X_{+-}(T) = \mathbf{E}_{+,T}^{-*} \Gamma_{\text{pp},+-}(T).$$

The singular part of this cross-block is, in logarithmic coordinate, $\int_0^T q_{+}(T-t)^{*} f_{-}(t) dt$. There are exactly two roots on each side. The Euclidean norms of these two rows are bounded by $\sqrt{2}e^{-\delta(T-t)}P_m(T-t)$ and $\sqrt{2}e^{-\delta t}P_m(t)$, respectively. Hence its norm is at most

$$2e^{-\delta T} \int_0^T P_m(T-t)P_m(t)dt = 2e^{-\delta T}V_m(T).$$

The equality follows by integrating each monomial: $\int_0^T (T-t)^j t^l / (j!l!) dt = T^{j+l+1}/(j+l+1)!$, as follows by j integrations by parts, or the beta integral proved in Lemma 2. For the three regular contributions to the untransformed cross-block, absolute integration and (EG23) give the uniform bound C_{+-} , by the estimates used for (EG32). Multiplication by $\mathbf{E}_{+,T}^{-*}$ multiplies this bound by at most $e^{-\delta T}P_m(T)$. Therefore $\|X_{+-}(T)\| \leq \eta_{+-}(T)$, with no cancellation assumed.

The left cutoff matrices increase in the positive-semidefinite ordering as T increases. By the determinant–trace argument in (EG36), $\Gamma_{\text{pp},--}(T) \succeq g_{-}I$ for $T \geq 1$. Its inverse has norm at most g_{-}^{-1} , so the entire transformed correction $X_{+-}(T)\Gamma_{\text{pp},--}(T)^{-1}X_{+-}(T)^{*}$ has norm at most $\eta_{+-}(T)^2/g_{-}$. The triangle inequality proves (EG39c). The degree of V_m is $2m-1$, whereas that of P_m is $m-1$. Applying the relative determinant estimate from Theorem 4 gives the asserted $e^{-2\delta T}(1+T)^{4m-2}$ remainder. The complete left minimum, its fixed positive lower bound, and all cross terms have been retained in the original quotient. $\square$

An exact finite formula is available even for the cross-block singular terms. For $\alpha = \bar{\rho} + \sigma - 1$ and $n = i + j$, their entry is $I_n(\alpha, T)/(i!j!)$, where

$$I_n(\alpha, T) = \begin{cases} e^{\alpha T} \sum_{k=0}^n \frac{(-1)^k n! T^{n-k}}{(n-k)! \alpha^{k+1}} - \frac{(-1)^n n!}{\alpha^{n+1}}, & \alpha \neq 0, \\ T^{n+1}/(n+1), & \alpha = 0. \end{cases} \quad (\text{EG41})$$

Differentiation in T and the value zero at $T = 0$ verify this formula directly. The finite regular contributions are the integrals in (EG28) and (EG29) with the appropriate rows. Thus every finite cross term is explicitly retained, including the resonant $\alpha = 0$ terms.

12 All nilpotent singular scales

Theorem 5 (Every scale of one right primary). *For one right primary of full multiplicity m , put $b_\rho = 2\Re\rho - 1$. The decreasing eigenvalues of the actual $\Gamma_{\text{pp}}(T)$ satisfy*

$$\lambda_r(T) = e^{b_\rho T} \left(\frac{(r-1)!}{(m-r)!} \right)^2 b_\rho^{-(2r-1)} T^{2(m+1-2r)} (1 + O_{\rho,m}(T^{-1})), \quad 1 \leq r \leq m. \quad (\text{EG42})$$

They are exactly the generalized eigenvalues of the original pair $(\Gamma_\rho(T), \mathbf{U}_\rho^ \mathbf{U}_\rho)$.*

Proof. Write $P(T) = e^{TJ_m^\dagger}$. Its entry (i, j) , with indices starting at zero, is $T^{j-i}/(j-i)!$ if $j \geq i$ and zero otherwise. Every nonzero minor of order r is homogeneous of degree $\sum J - \sum I$. The maximal possible degree is $r(m-r)$, attained uniquely by $I = \{0, \dots, r-1\}$ and $J = \{m-r, \dots, m-1\}$. Its coefficient is

$$c_{m,r} = \det \left[\frac{1}{(m-r+j-i)!} \right]_{i,j=0}^{r-1} = \prod_{j=0}^{r-1} \frac{j!}{(m-r+j)!}. \quad (\text{EG43})$$

A negative factorial in the determinant denotes the zero entry, as in P . To prove the equality, multiply column j by $(m-r+j)!$. Entry (i, j) becomes the falling factorial $(m-r+j)_i$, a monic polynomial of degree i in the column argument. The determinant of these polynomials is the Vandermonde of the consecutive arguments, equal to $\prod_{j=0}^{r-1} j!$. Dividing back proves (EG43).

The matrix of $\bigwedge^r P(T)$ consists of those minors. After division by $T^{r(m-r)}$ it equals $c_{m,r}|I\rangle\langle J| + O(T^{-1})$ in operator norm. Therefore, on taking its Gram with $\bigwedge^r \widehat{\mathbf{G}}$,

$$T^{-2r(m-r)} \bigwedge^r (P(T)^* \widehat{\mathbf{G}} P(T)) = c_{m,r}^2 \det(\widehat{\mathbf{G}}_{I,I}) |J\rangle\langle J| + O(T^{-1}).$$

The determinant of this leading r -primary Gram is $b_\rho^{-r^2}$ by (EG16). The largest eigenvalue of an exterior power of a positive matrix is the product of its r largest eigenvalues: diagonalizing the positive matrix gives all its exterior eigenvectors as wedges, with the corresponding products. Hence that product is

$$e^{rb_\rho T} c_{m,r}^2 b_\rho^{-r^2} T^{2r(m-r)} (1 + O(T^{-1})).$$

For $r = m$ the exterior identity is exact, and the same conclusion holds. The relative metric error in (EG31) is exponentially small, so the min-max principle preserves these products with factor $1 + o(T^{-1})$. Explicitly the error yields $(1 - \varepsilon_T) \mathbf{E}_T^* \widehat{\mathbf{G}} \mathbf{E}_T \preceq \Gamma_{\text{pp}}(T) \preceq (1 + \varepsilon_T) \mathbf{E}_T^* \widehat{\mathbf{G}} \mathbf{E}_T$, and applying the Rayleigh min-max formula bounds every eigenvalue by the same two factors. Consecutive products give $c_{m,r}/c_{m,r-1} = (r-1)!/(m-r)!$, proving (EG42). Finally substituting $a = \mathbf{U}_\rho u$ in the two quadratic forms changes the pair to $(\Gamma_{\text{pp}}(T), I)$, which proves the generalized-eigenvalue assertion without an isometry claim. $\square$

13 Critical primary: the exact finite metric and stronger error

For one original critical primary $\rho = 1/2 + i\gamma$ of full multiplicity m , put

$$D_T = \text{diag}(T^{j+1/2})_{j=0}^{m-1}, \quad (H_m)_{ij} = \frac{1}{i!j!(i+j+1)}. \quad (\text{EG44})$$

The matrix F_ρ is still the full convergent integral (EG28) before subtraction of $\widehat{\mathbf{G}}$; a right Cauchy Gram is not defined at this critical root.

Theorem 6 (Critical determinant, including its first finite correction). *For $T \geq 1$,*

$$\Gamma_{\text{pp}}(T) = D_T H_m D_T + F_\rho - B_\rho^{\text{reg}}(T), \quad D_T^{-1} \Gamma_{\text{pp}}(T) D_T^{-1} = H_m + O_\rho(T^{-1}). \quad (\text{EG45})$$

More precisely, if $f_{00} = (F_\rho)_{00}$ in the original principal-part coordinates, then

$$\begin{aligned} \log \det \Gamma_\rho(T) &= m^2 \log T - 2m \log \left(2 \left| \frac{\zeta^{(m)}(\rho)}{m!} \right| \right) + \log \prod_{j=0}^{m-1} \frac{j!}{(m+j)!} \\ &\quad + \frac{m^2 f_{00}}{T} + O_\rho(T^{-2}). \end{aligned} \quad (\text{EG46})$$

In particular the supplied (55)–(56) hold, with stronger remainders.

Proof. The oscillatory scalar $e^{i\gamma t}$ cancels in each single-primary Gram entry. Thus the singular integral is exactly $T^{i+j+1}/(i!j!(i+j+1))$, giving the first identity in (EG45). The regular integrals converge by the proof of Theorem 3, since $\beta = 1/2 < 3$; their tail has the explicit bound (EG33). Since $\|D_T^{-1}\| = T^{-1/2}$, the transformed regular term is $O(T^{-1})$. Entrywise it is more precisely

$$D_T^{-1}(F_\rho - B_\rho^{\text{reg}}(T))D_T^{-1} = \frac{f_{00}}{T} e_0 e_0^* + O(T^{-2}). \quad (\text{EG47})$$

The determinant of D_T squared is $T^{\sum_{j=0}^{m-1} (2j+1)} = T^{m^2}$. The ordinary Cauchy formula applied to $1/(i+j+1)$ and then division by $\prod (i!)^2$ gives $\det H_m = \prod_{j=0}^{m-1} j!/(m+j)!$. The identity $(H_m^{-1})_{00} = m^2$ is proved in Lemma 2 below. Apply the matrix logarithm expansion used in Theorem 4 to (EG47); its first trace is $m^2 f_{00}/T$ and its remainder is $O(T^{-2})$. The original $|\det U_\rho|^2$ supplies the unchanged local zeta factor, proving (EG46). $\square$

Lemma 2 (The three endpoint kernel values, with proof). *Let $b_j = 1/j!$ and $e_0 = (1, 0, \dots, 0)^T$. Then*

$$b^* H_m^{-1} b = e_0^* H_m^{-1} e_0 = m^2, \quad e_0^* H_m^{-1} b = (-1)^{m-1} m. \quad (\text{EG48})$$

Proof. The matrix H_m is the Gram of $v_j(x) = x^j/j!$ on $[0, 1]$. Use the explicit shifted Legendre polynomials

$$L_n(x) = \frac{1}{n!} \frac{d^n}{dx^n} \{x^n (x-1)^n\}.$$

This is the Rodrigues formula, also recorded in [3, §18.5(ii)]. Integration by parts n times shows L_n is orthogonal to every polynomial of degree $< n$: the derivatives of $x^n(x-1)^n$ through order $n-1$ vanish at both endpoints. Its leading coefficient is $(2n)!/(n!)^2$. The same integration by parts gives

$$\int_0^1 L_n(x) x^n dx = \int_0^1 x^n (1-x)^n dx = \frac{(n!)^2}{(2n+1)!}.$$

The last identity follows by integrating the beta integral by parts successively, starting with $\int_0^1 x^q dx = 1/(q+1)$. Therefore $\int_0^1 L_n^2 = 1/(2n+1)$. Expanding at $x = 0, 1$ in the Rodrigues formula gives $L_n(0) = (-1)^n$, $L_n(1) = 1$. It follows that $\sqrt{2n+1} L_n$, $0 \leq n < m$, is an orthonormal basis of the degree- $< m$ polynomials. Its evaluation kernel is

$$K_m(x, y) = \sum_{n=0}^{m-1} (2n+1) L_n(x) L_n(y) = v(x)^T H_m^{-1} v(y).$$

The second equality follows because either side reproduces every polynomial under integration against $v(y)$; the Gram is positive definite. At the endpoints, $v(0) = e_0$, $v(1) = b$. The diagonal sum is $\sum (2n+1) = m^2$, and the alternating sum is $\sum (-1)^n (2n+1) = (-1)^{m-1} m$, by induction. This proves all three identities without assuming endpoint kernel values. $\square$

14 Relative arithmetic weight and its typed boundary map

Theorem 7 (Original relative arithmetic spectra). *Write $W_Z(T) = A_Z^* \Gamma_Z(T) + \Gamma_Z(T) A_Z - \Gamma_Z(T)$. For a right divisor the spectrum of $\Gamma_Z(T)^{-1/2} W_Z(T) \Gamma_Z(T)^{-1/2}$ tends to*

$$\left\{ \sum_{\rho} m_{\rho} (2\Re \rho - 1), 0, \dots, 0 \right\}. \quad (\text{EG49})$$

For a single critical primary of multiplicity $m \geq 2$, its two leading eigenvalues are

$$\frac{\pm m \sqrt{m^2 - 1}}{T} + O_{\rho}(T^{-2}), \quad (\text{EG50})$$

and all its remaining eigenvalues are $O_{\rho}(T^{-2})$. For $m = 1$ the defect is exactly zero at every finite T .

Proof. A simultaneous invertible congruence of a Hermitian pair preserves its generalized eigenvalues, because $\det(S^*(W - \lambda \Gamma)S) = |\det S|^2 \det(W - \lambda \Gamma)$. Use $S = U_Z^{-1} E_T^{-1}$. By (EG8) and $E_T A_{\text{pp}} = A_{\text{pp}} E_T$, the action in these coordinates is still A_{pp} , while the metric tends to $\widehat{G}$. Conjugating (EG17) gives

$$A_{\text{pp}}^* \widehat{G} + \widehat{G} A_{\text{pp}} - \widehat{G} = e e^*.$$

Its relative Hermitian matrix has one eigenvalue $e^* \widehat{G}^{-1} e = \sum m_{\rho} (2\Re \rho - 1)$ by (EG19), and all others zero. The positive definite inverse square root is continuous here: conjugate the metric by $\widehat{G}^{-1/2}$ to obtain $I + X_T$, whose inverse square root has its norm-convergent binomial series for $\|X_T\| < 1$. Thus the relative matrices converge in norm; the Rayleigh min-max formula gives convergence of their ordered eigenvalues. This proves (EG49).

For the critical block, use $S = U_{\rho}^{-1} D_T^{-1}$. The action becomes $D_T A_{\text{pp}} D_T^{-1} = \rho I + N/T$, $N = J_m^{\uparrow}$. The real scalar part cancels the weight one. The metric is $H_m + O(T^{-1})$ by (EG45). Hence the relative defect is T^{-1} times the relative form of $N^* H_m + H_m N$, plus $O(T^{-2})$ in operator norm. Entrywise,

$$N^* H_m + H_m N = b b^* - e_0 e_0^*.$$

Indeed the (i, j) entry for $i + j > 0$ is $(i + j)/(i! j! (i + j)) = 1/(i! j!)$, and the $(0, 0)$ entry is zero. Let $v = H_m^{-1/2} b$ and $w = H_m^{-1/2} e_0$. Lemma 2 gives $\|v\|^2 = \|w\|^2 = m^2$, $w^* v = (-1)^{m-1} m$. The nonzero eigenvalues of $vv^* - ww^*$ sum to zero and their squared sum is $2(m^4 - m^2)$, by directly taking the trace of its square. For $m \geq 2$ the vectors are independent, so its rank is two and its eigenvalues are exactly $\pm m \sqrt{m^2 - 1}$. The min-max bound proves (EG50). If $m = 1$, $A_Z = \rho I$ and $2\Re \rho - 1 = 0$, giving exact vanishing for every metric, including the actual finite cutoff metric. $\square$

There is also a finite bound for the convergence in (EG49). Set $\Omega_T = E_T^{-*} \Gamma_{\text{pp}}(T) E_T^{-1}$, $X_T = \widehat{G}^{-1/2} (\Omega_T - \widehat{G}) \widehat{G}^{-1/2}$, $B = \widehat{G}^{1/2} A_{\text{pp}} \widehat{G}^{-1/2}$ and $L_0 = B^* + B - I$. If $\delta_T = \|X_T\| < 1$, the transformed relative matrix differs from L_0 by at most

$$\frac{\delta_T}{1 - \delta_T} (\|L_0\| + 2\|B\| + 1). \quad (\text{EG51})$$

To verify this bound, write it as $(I + X_T)^{-1/2} (L_0 + B^* X_T + X_T B - X_T) (I + X_T)^{-1/2}$. The two factors have norm at most $(1 - \delta_T)^{-1/2}$ and their difference from I has norm at most $(1 - \delta_T)^{-1/2} - 1$. Expanding the first term around L_0 gives $\|L_0\| \delta_T / (1 - \delta_T)$; the other terms give the rest. Inserting (EG31) divided by g_* makes this a fully finite source estimate.

Theorem 8 (Typed Beurling/source comparison and exact boundary). *Suppose the right divisor is stable under conjugation, with the same full multiplicities. Let C be the linear permutation $(Ca)_{\rho,j} = a_{\bar{\rho},j}$; it does not conjugate coefficients. Then*

$$C^* = C = C^{-1}, \quad C^*GC = \widehat{G}, \quad CA_{pp} = A_Z^*C. \quad (\text{EG52})$$

The invertible map from original source jets to Beurling quotient jets is

$$M_T = GCE_TU_Z, \quad M_T^{-1} = U_Z^{-1}E_T^{-1}CG^{-1}. \quad (\text{EG53})$$

It satisfies the two exact identities

$$M_T^*H_B^ZM_T = U_Z^*E_T^*\widehat{G}E_TU_Z, \quad (\text{EG54})$$

$$M_TA_Z = (I - A_Z)M_T + e\ell_T, \quad \ell_T = e^*CE_TU_Z = e^*E_TU_Z. \quad (\text{EG55})$$

Here ℓ_T is exactly $e^{-T/2}$ times the singular part of $\psi_u(e^{-T})$ as a functional of u , and is nonzero.

Proof. The first two claims in (EG52) follow by substituting the indices in (EG10). For the action, the block of CA_{pp} at output ρ has diagonal $\bar{\rho}$ and the upper shift, exactly as in A_Z^*C . This verifies domains, codomains, complex conjugates, and shift directions. Every factor in (EG53) is invertible. Since $H_B^Z = G^{-1}$, (EG54) follows by direct multiplication. For the action, (EG8), commutation with E_T , and (EG52) give

$$M_TA_Z = GA_Z^*CE_TU_Z = \{(I - A_Z)G + ee^*\}CE_TU_Z,$$

where the last equality is exactly (EG17). This proves (EG55). The permutation fixes e , so the two formulas for ℓ_T agree. Its evaluation is

$$\ell_T u = \sum_{\rho,j} (U_Z u)_{\rho,j} e^{(\rho-1/2)T} \frac{T^j}{j!},$$

the stated original endpoint value. It is nonzero since $e \neq 0$ and E_TU_Z is invertible. This is an exact rank-one boundary equation, not an unqualified action intertwining assertion. $\square$

Theorem 9 (The full physical integration-by-parts identity). *Define the original highest-coefficient row by $\ell_{\text{ref}}(u) = [s^{d-1}]r_Z(u)(s)$, and let $\Psi(x)u = \psi_u(x)$. Then*

$$D\Psi = \Psi A_Z + \phi_*\ell_{\text{ref}}. \quad (\text{EG56})$$

At every finite T , with $a_T = e^{-T}$,

$$\begin{aligned} W_Z(T) &= a_T \Psi(a_T)^* \Psi(a_T) \\ &\quad - \ell_{\text{ref}}^* \int_{a_T}^{\infty} \overline{\phi_*(x)} \Psi(x) dx - \left(\int_{a_T}^{\infty} \Psi(x)^* \phi_*(x) dx \right) \ell_{\text{ref}}. \end{aligned} \quad (\text{EG57})$$

The formulas hold with the original nonisometric maps and both finite source cross terms.

Proof. Because r_Z is multiplication by the original unit modulo h_Z , the polynomial $sr_Z(u) - r_Z(A_Z u)$ is divisible by h_Z . Its degree is at most d ; h_Z is monic and the leading coefficient is $\ell_{\text{ref}}(u)$. Hence it is exactly $h_Z \ell_{\text{ref}}(u)$. Multiplication by Φ/h_Z in (EG5) and integration by parts for D prove (EG56). The boundary terms for that Mellin calculation vanish on $\Re s > 1$ by (EG9) and the Gaussian tail, so the function identity follows by uniqueness.

At the physical cutoff $a_T > 0$ all functions are smooth and the upper boundary terms vanish. Direct differentiation gives

$$\int_{a_T}^{\infty} \{(D\Psi)^*\Psi + \Psi^*(D\Psi) - \Psi^*\Psi\} dx = - \int_{a_T}^{\infty} \frac{d}{dx} (x\Psi^*\Psi) dx = a_T \Psi(a_T)^* \Psi(a_T).$$

Substituting (EG56) gives (EG57), including both negative cross terms. No quotient or endpoint limit has been applied inside this finite identity. The convergence and spectra in Theorem 7 follow from the proved exact source metric, and do not discard these finite terms. $\square$

15 Equation correspondence and provenance

The input is the final mathematical response in the supplied *ChatGPT-Research Program Audit-20260919-2248.md*, sections 5–7, together with the original source proof *EX-ACT_CORPUS_RECEIVERS.tex*, (B1)–(B5), (T2), (MI5)–(MI9). The original pasted input is retained verbatim beside this proof. The following are proof locators, not replacements for the proofs above.

Supplied equations	Exact proof in this document
(32)–(38)	Section 8, Theorem 1, (EG14)–(EG15)
(39)–(42)	Theorem 2
(43)–(44), (57)	Proposition 1
(45)–(50)	Theorem 3, (EG34)–(EG35)
(51)	Lemma 1
(52)–(53)	Theorem 4, Proposition 2
(54)	Theorem 5
(55)–(56)	Theorem 6
(58)–(59)	Theorem 7, Lemma 2
(60)–(61)	Theorem 8
(62)	Theorem 9

The input’s displayed formulas are valid on their stated domains. For a signed choice of the pair height the logarithm in (53) requires $|\gamma|$; its stipulated positive height already satisfies this. The new finite correction (EG28)–(EG33) strengthens the endpoint error and the determinant remainder. The actual critical remainder improves (55)–(56) to (EG45)–(EG47), evaluates its first finite determinant correction, and improves the relative defect error from $O(T^{-3/2})$ to $O(T^{-2})$. The full quartet minimum has its own proved finite bound (EG39a)–(EG39c), giving the determinant remainder $O_Z(e^{-2\delta T}(1+T)^{4m-2})$; the stronger unminimized right-block rate is not asserted for that different finite Schur correction.

References

- [1] R. Roy, F. W. J. Olver, R. A. Askey, R. Wong, and W. P. Reinhardt, NIST Digital Library of Mathematical Functions, §1.3(ii), equation (1.3.14), Cauchy determinant, <https://dlmf.nist.gov/1.3.E14>. Exact determinant identity; its confluence and all multiplicity factors are proved above.
- [2] R. Roy, F. W. J. Olver, R. A. Askey, R. Wong, and W. P. Reinhardt, NIST Digital Library of Mathematical Functions, §1.14, Integral Transforms, Fourier and Mellin transform conventions and Parseval formulas, <https://dlmf.nist.gov/1.14>.
- [3] T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, NIST Digital Library of Mathematical Functions, §18.5(ii), Rodrigues formulas, <https://dlmf.nist.gov/18.5.ii>. The shifted formula, orthogonality, norms, and endpoint evaluations used here are proved in full above.
- [4] Jean-François Burnol, *A lower bound in an approximation problem involving the zeros of the Riemann zeta function*, arXiv:math/0103058, <https://arxiv.org/abs/math/0103058>. Historical finite-zero projection context only; no unproved approximation claim is imported.